Trøndelagkonkurranse 2026 - Level II

Third Round - October 31, 2026

1st Part

Choose and solve only one of the following problems:

Problem 1

Let $A = \{1, 3, 4\}$. We can take any two different elements of A, say a and b, and replace them with c and d such that

$$c = \frac{3a - 4b}{5} \quad \text{and} \quad d = \frac{4a + 3b}{5}$$

Let $B = \{2, 3, 5\}$. Determine whether A can be transformed into B through successive transformations.

Problem 2

A natural number $n > 1$ is called *agreeable* if an equilateral triangle T that can be divided into n smaller equilateral triangles such that:

- The sides of the smaller triangles are parallel to the sides of T.
- The smaller triangles do not overlap inside T. Vertices and edges can touch each other.
- Every portion of T is covered by a smaller triangle.

Show that 2026 is agreeable.